Cross-Platform Price Variation Analysis of Maybelline Products

1. Introduction

The rapid growth of e-commerce platforms has led to increased competition in pricing strategies across different platforms. Consumers today often encounter variations in the price of identical products depending on the platform they choose. This study focuses on analyzing such variations for selected Maybelline products across multiple platforms and understanding the role of discounts in influencing final prices.

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2. Objective

The objectives of this study are:

- To compare product pricing across multiple e-commerce platforms
- To analyze discount strategies adopted by different platforms
- To study the relationship between discount percentage and final price
- To identify price variations for identical products

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3. Data Collection and Preparation

Data for this study was collected manually from multiple e-commerce platforms including Amazon, Flipkart, Nykaa, Myntra, Purplle, and the brand's official platform.

3.1 Product Selection

A set of commonly available Maybelline products was selected across three major categories:

- Lipstick
- Foundation
- Mascara

Only those products that were consistently available across multiple platforms were included to ensure comparability.

3.2 Handling Product Variants

Each product typically exists in multiple variants such as shades, sizes, or formulations. These variants often have slight differences in pricing.

To address this:

- Multiple variants of the same product were considered during data collection
- Price, discount, and final price values were recorded for these variants

To avoid bias from any single variant:

- **Aggregated values (average metrics)** were calculated
- This ensured that **no individual shade or variant disproportionately influenced the results**

This approach provided a more representative estimate of pricing for each product.

3.3 Data Cleaning and Standardization

To ensure consistency and reliability of the dataset:

- All prices were standardized in Indian Rupees (₹)
- Discount percentages were converted into a uniform format
- Duplicate or inconsistent entries were removed
- Only valid and comparable product-platform combinations were retained

This step ensured that the dataset was clean, structured, and suitable for analysis.

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4. Methodology

The analysis was conducted using Microsoft Power BI. The methodology involved multiple stages:

4.1 Data Transformation

The cleaned dataset was imported into Power BI, where:

- Key variables such as **Average MRP**, **Average Discount**, and **Average Final Price** were computed
- Aggregated measures were used instead of raw values to maintain consistency

4.2 Data Modeling

The dataset was structured in a tabular format with the following key attributes:

- Product Name
- Category
- Platform
- Average MRP (₹)
- Average Discount (%)
- Average Final Price (₹)

This structure enabled efficient filtering and comparison across different dimensions.

4.3 Visualization Design

Multiple visualizations were created to represent different aspects of the data:

- **Bar charts** to compare pricing across platforms
- **Clustered charts** to analyze MRP vs final price
- **Scatter plots** to study the relationship between discount and final price
- **Product-level charts** to compare identical products across platforms

Interactive slicers were added for:

- Category
- Product Name

This allowed dynamic filtering and improved user interaction.

4.4 Analytical Approach

The analysis focused on identifying patterns such as:

- Price differences for the same product across platforms
- Variation in discount strategies
- Relationship between discount levels and final prices

Special attention was given to ensure that:

- Aggregated metrics were used consistently
- Comparisons were made on identical products

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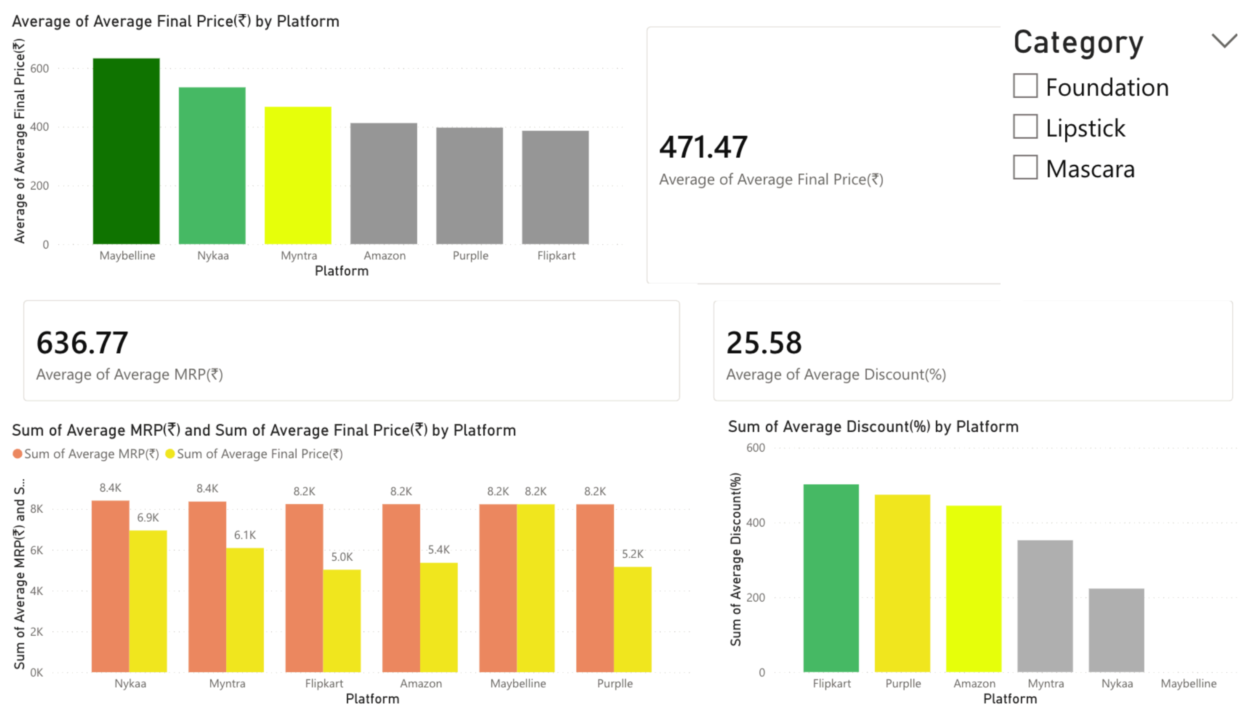
5. Analysis and Findings

5.1 Platform-Level Pricing & Discount Overview



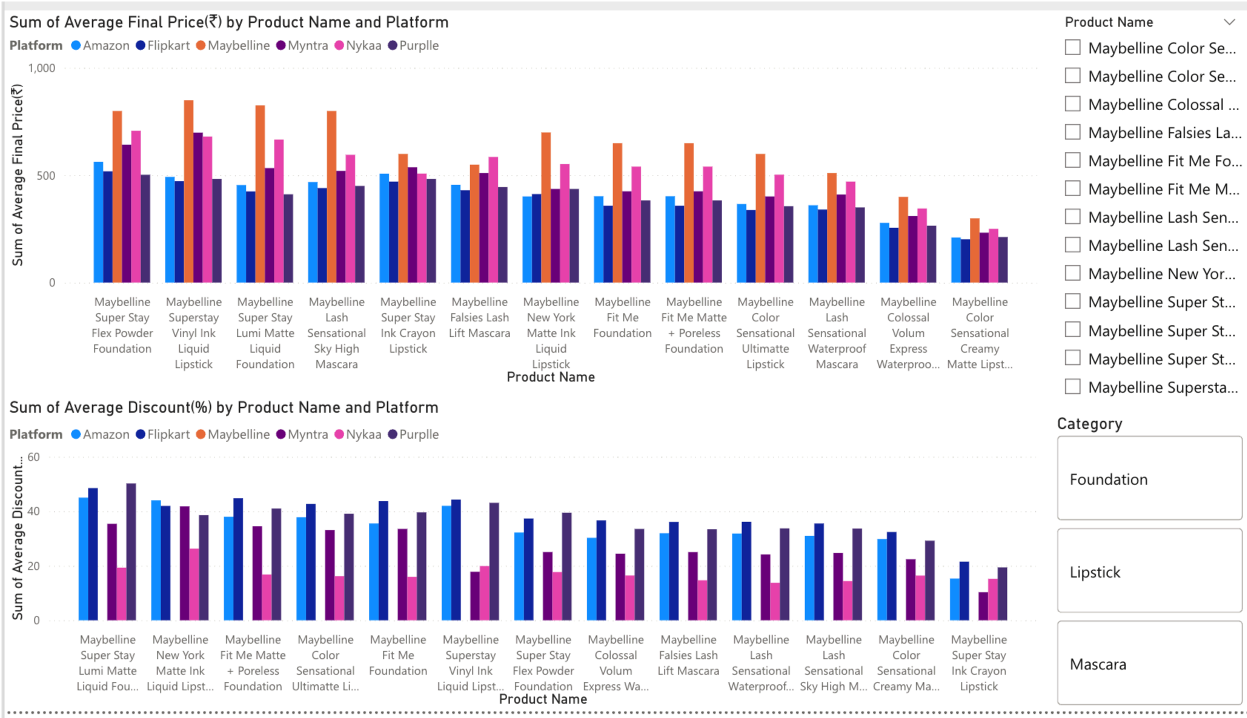
The first dashboard provides an overview of pricing and discount strategies across different platforms. It compares the average final price, average MRP, and average discount percentages.

From the analysis, it is evident that there are clear differences in pricing strategies across platforms. Platforms such as Amazon and Flipkart tend to offer higher discounts, which results in lower final prices for consumers. In contrast, platforms like Nykaa maintain relatively higher final prices with comparatively lower discounts, indicating a more premium pricing strategy.

The comparison between MRP and final price highlights the impact of discounts. Platforms with a larger gap between MRP and final price are using aggressive promotional strategies, while others maintain stable pricing with smaller reductions.

Overall, this dashboard establishes that pricing is not standardized across platforms and is heavily influenced by discount strategies.

5.2 Product-Level Price Comparison



The second dashboard focuses on product-level analysis, comparing the prices of identical Maybelline products across different platforms.

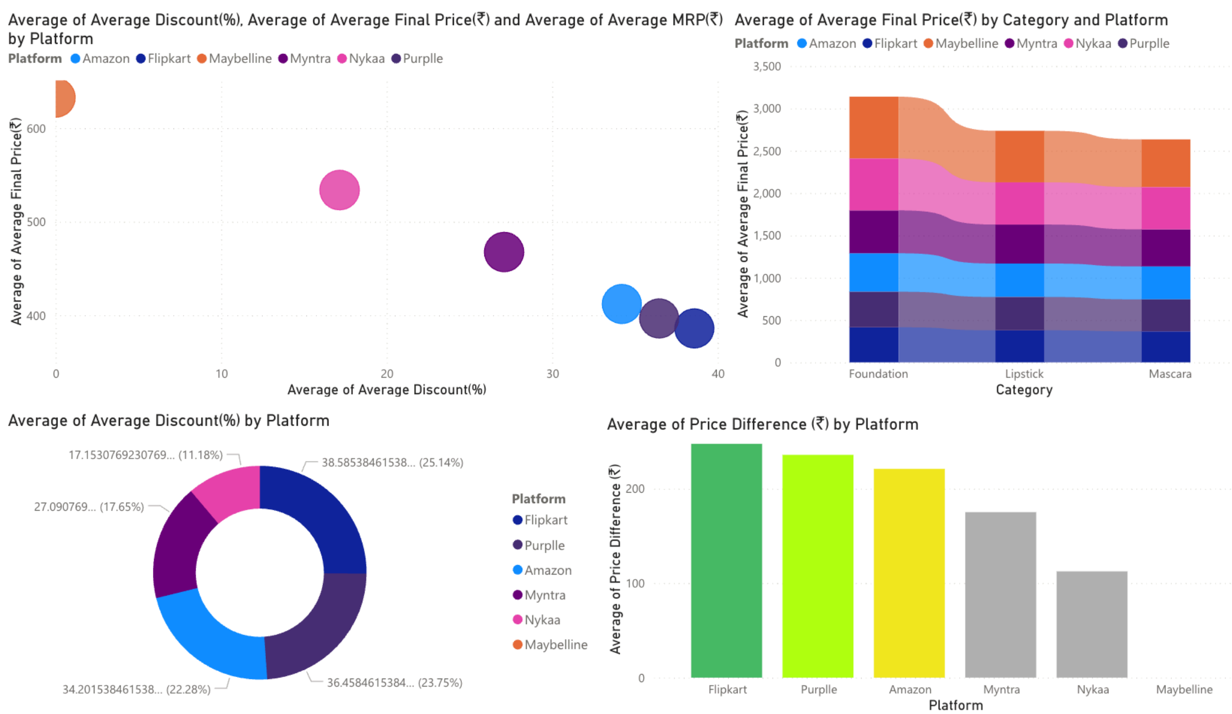
This analysis clearly shows that the same product is sold at different prices depending on the platform. The variation is consistent across multiple products, indicating the presence of price dispersion in the market.

Additionally, the discount comparison at the product level reveals that some platforms consistently offer higher discounts across multiple products, while others follow a more stable pricing approach.

The use of slicers allows filtering by category and product, making it possible to observe how price variation differs across product types such as lipsticks, foundations, and mascaras.

This dashboard highlights the importance of comparing platforms before making a purchase, as consumers can benefit from significant price differences.

5.3 Discount–Price Relationship and Strategic Insights



The third dashboard focuses on understanding the relationship between discount percentages and final prices using a scatter plot and supporting visuals.

The scatter plot shows a clear inverse relationship between discount and final price. Platforms offering higher discounts are associated with lower final prices, confirming the direct impact of discount strategies on consumer cost.

The size of the bubbles, representing MRP, further indicates how heavily discounted products achieve lower final prices despite having higher base prices.

Additional visuals, such as price difference comparisons, highlight the actual savings provided by each platform. Platforms with higher price differences indicate stronger discounting strategies.

Overall, this dashboard provides a deeper analytical perspective by linking pricing, discounting, and platform positioning, reinforcing the conclusion that discount strategies play a crucial role in competitive pricing.

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6. Key Insights

- Pricing strategies differ significantly across platforms
- Discounts play a crucial role in determining final prices
- Identical products are priced differently across platforms
- Consumers can benefit from cross-platform comparison

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7. Conclusion

This study highlights the presence of price variation across e-commerce platforms and emphasizes the role of discount strategies in shaping final prices. The use of aggregated metrics ensured a balanced and unbiased comparison across product variants, strengthening the reliability of the findings.

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8. Limitations

- Limited product sample size
- Data collected at a specific point in time
- Dynamic pricing may cause variations over time

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9. Tools Used

- Microsoft Power BI
- Microsoft Excel
